Ke'Anna McKinney

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Professional Summary

Detail-oriented digital forensics student with hands-on experience in forensic acquisition, chain-of-custody documentation, hash-based integrity verification, and reproducible case documentation. Proficient with industry-standard tools including FTK Imager, KAPE, Volatility 3, Autopsy, and Wireshark across Windows and Linux environments. Practiced in following structured forensic workflows, maintaining complete and auditable work papers, and communicating findings clearly to both technical and non-technical audiences. Committed to ethical evidence handling and precise, defensible documentation in every investigation.

Education

Sam Houston State University
Master of Science in Digital Forensics. *Expected December 2027*
Relevant Coursework: Ethical Hacking, Digital Forensics Investigation, Network and Cyber Security, Principle & Policy - Info Assurance, Organization System Security, Malware, Responsible AI

Louisiana State University Baton Rouge, LA
Bachelor of Science in Computer Science, Concentration in Cybersecurity. *May 2026*
Relevant Coursework: Software Vulnerabilities & Exploitation, Digital Forensics, Networking Fundamentals, Operating Systems

Skills

Digital Forensics Tools: FTK Imager, KAPE, Volatility 3, Autopsy, Impacket, John the Ripper, Wireshark

Evidence Handling & Documentation: Chain-of-Custody Documentation, Hash-Based Integrity Verification, NIST SP 800-61, Reproducible Case Documentation, Git Version Control

Systems & Platforms: Windows & Linux Administration, Active Directory, Microsoft 365, Google Workspace, Microsoft OneDrive, Google Drive

Programming & Data: Python, SQL, Bash, Data Collection & Analysis

Tools: Microsoft Office Suite (Word, Excel, PowerPoint), Splunk, VMware, VirtualBox

Experience

Undergraduate Research Assistant 2024 – 2026
SysSec Lab, Louisiana State University *Baton Rouge, LA*

- Conducted a compliance and privacy gap analysis examining whether mobile applications disclosed third-party data-sharing practices in their privacy policies, assessing risk across 1,000+ data sources including app network traffic, API data calls, and app store metadata.
- Built Python automation scripts to collect, organize, and analyze technical data, maintaining version-controlled documentation using Git to keep all records auditable and reproducible.

Student Success Mentor November 2024 – May 2026
Louisiana State University *Baton Rouge, LA*

- Supported students with academic and administrative questions, coordinating cross-functionally with faculty and support teams and maintaining detailed documentation of interactions and follow-up actions.
- Managed multiple concurrent student cases independently while collaborating with a diverse team of faculty and staff to resolve issues.

Projects

DFIR Endpoint Intrusion Investigation August 2026
Simulated Enterprise Sandbox

- Investigated a simulated Windows 10 endpoint compromise, acquiring live volatile memory with FTK Imager and extracting triage artifacts with KAPE prior to host isolation, following chain-of-custody and integrity-verification procedures throughout.
- Performed memory forensics using Volatility 3 to reconstruct the process execution tree, identifying a macro-based phishing payload that spawned PowerShell and established command-and-control communication.
- Analyzed registry and prefetch artifacts to confirm persistence via a Registry Run Key, correlated execution timestamps against the Master File Table, and ingested cataloged indicators of compromise into Splunk for analysis.

Active Directory & Windows Server Home Lab July 2026
Microsoft Azure

- Provisioned and secured a Windows Server 2022 domain controller in Azure (Trusted Launch, Secure Boot/vTPM), configuring a static private IP and Azure Bastion for encrypted remote access with zero public inbound ports exposed.
- Installed Active Directory Domain Services, DNS, and DHCP roles, promoted the server, and established a new AD forest and domain with a two-branch OU structure containing dedicated Users, Computers, and Groups containers.

- Diagnosed and resolved a failed client domain join by identifying a DNS misconfiguration, verifying the fix via command-line diagnostics and confirming the new computer object in Active Directory.

Security Configuration & Vulnerability Assessment August 2026
Kali Linux / Metasploitable Lab
- Performed network reconnaissance and vulnerability scanning against an isolated Metasploitable target using Nmap service/version detection and NSE scripts, identifying 23 exposed services and multiple critical CVEs including a vsFTPd 2.3.4 backdoor (CVE-2011-2523).
- Exploited the confirmed vsFTPd backdoor using Metasploit to obtain root-level remote command execution, documenting the full attack chain from discovery to compromise.
- Remediated four high-risk services (Telnet, an exposed root shell, UnrealIRCd, and the vulnerable FTP service) by modifying xinetd configurations, verifying remediation through before/after Nmap rescans.

Cybersecurity Risk Assessment Report August 2026
Simulated Financial Services Organization
- Conducted a simulated enterprise risk assessment across 10 organizational assets, including Active Directory, cloud storage, customer financial systems, and third-party GenAI tool usage, using a quantitative Likelihood x Impact risk-scoring methodology.
- Identified 3 critical and 7 high risks, including ransomware exposure on the file server and account takeover risk in Microsoft 365, and prioritized remediation based on calculated risk scores.
- Authored a formal risk register and assessment report mapping recommended controls to the NIST Cybersecurity Framework and 10 CIS Controls, supported by a companion Excel risk-tracking workbook.

Phishing Investigation, Incident Response Case Study August 2026
Self-Directed Simulation
- Investigated a simulated credential-harvesting phishing email targeting a fictional logistics company, following the SANS PICERL model cross-referenced against NIST SP 800-61.
- Performed email header analysis and validated SPF, DKIM, and DMARC authentication results to confirm sender spoofing and lookalike domain infrastructure.
- Extracted and cataloged indicators of compromise, mapped observed attacker behavior to five MITRE ATT&CK techniques, and authored a formal case study with containment, eradication, and recovery recommendations.

HIPAA Compliance Security Assessment August 2026
Healthcare Security Simulation
- Conducted a mock HIPAA Security Rule compliance assessment against a simulated clinic environment, evaluating access control, data handling, and audit logging against 45 CFR Sections 164.308 through 164.312.
- Performed technical vulnerability scanning with Nmap and configured centralized log forwarding to Splunk, validating an end-to-end audit logging pipeline that indexed 288 events.
- Identified 15+ high and critical severity vulnerabilities and produced a formal findings report with a control-by-control compliance matrix and prioritized remediation timeline.

TEAM PROJECTS

- HiTeK Forensics Challenge | Group Digital Forensics Investigation | 2026
- Collaborated with a 5-person team to investigate a simulated multi-system data exfiltration incident across Windows and Ubuntu disk images using Autopsy.
- Reconstructed a cross-system attack timeline by correlating bash history, authentication logs, and network capture evidence in Wireshark, tracing FTP-based file transfer and HTTP-based retrieval of exfiltrated data.
- Recovered user credentials from Windows SAM/SYSTEM hives and Linux shadow files using Impacket and John the Ripper, and co-authored a formal findings report detailing the who, what, when, and how of the incident.
- SOC Analyst Training | TryHackMe Blue Team Path & ReliaQuest SOC Bootcamp | 2025 - Ongoing
- Performed structured alert triage and log correlation across simulated security scenarios using Splunk SIEM, applying the 5 Ws framework and MITRE ATT&CK to produce clear, structured incident documentation.
- Produced post-incident investigation reports for both technical and non-technical stakeholders, translating complex technical findings into actionable conclusions.

CERTIFICATIONS

- CompTIA Security+ (In Progress, Expected 2026)
- Google Cybersecurity Professional Certificate (Completed)
- AWS Cloud Foundations (Completed)
- TryHackMe SOC Level 1 (In Progress)